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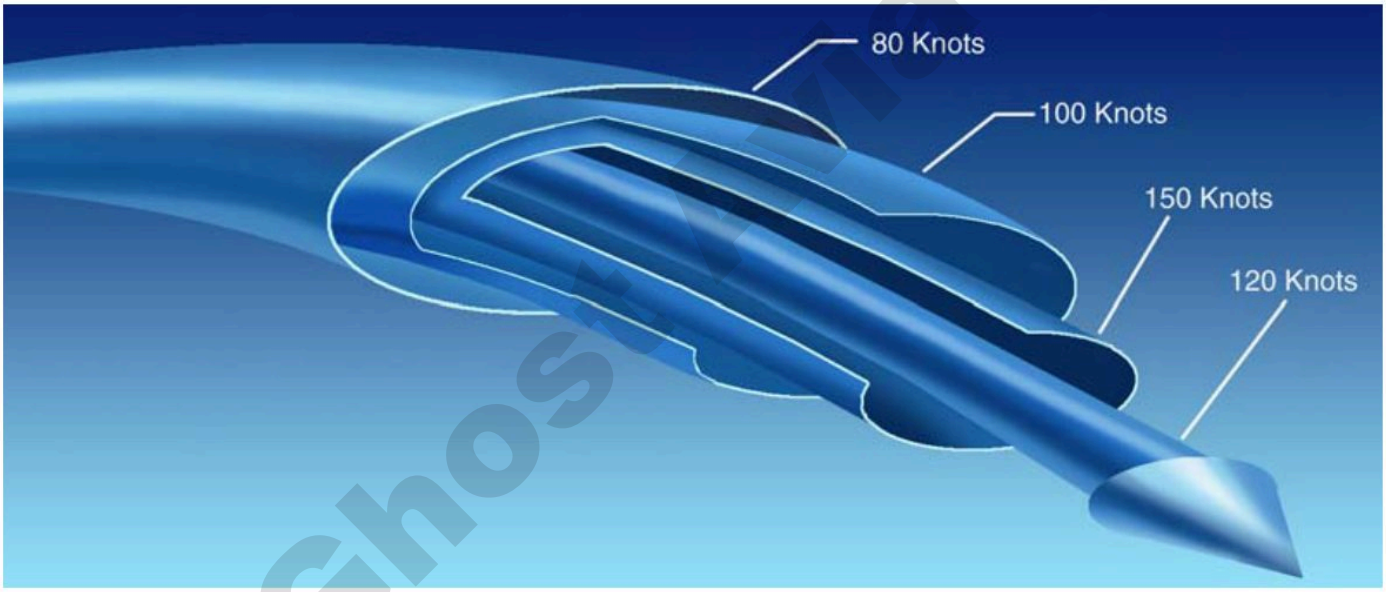
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1. Definition & Basic Properties

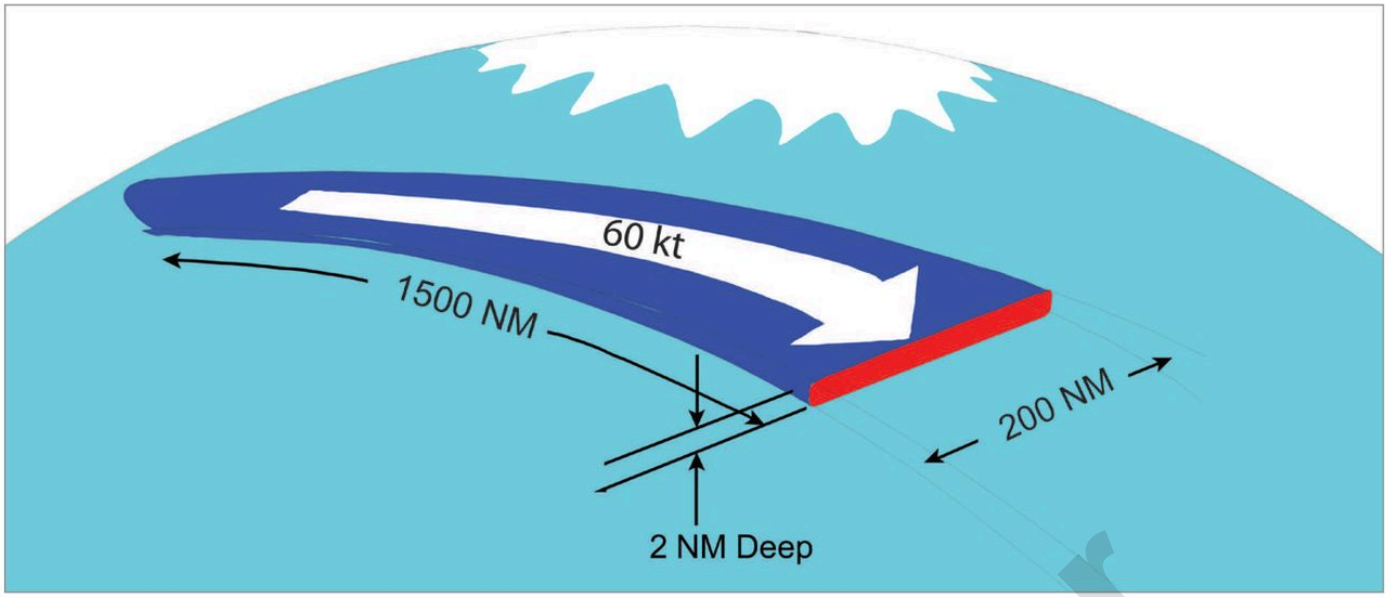
Jet Stream: A strong narrow current of winds along a semi-horizontal axis in the upper troposphere. Jet streams are present in both hemispheres all year round. The path of a jet stream is zigzag wavy like a serpentine.

- Length: thousands of km
- Width: hundreds of km
- Depth: some km (often less than **5 km**)
- **Vertical Wind Shear: about 5 m/s per km**
- **Arbitrary lower limit of jet core velocity: 30 m/sec (60 kt)**



Isotachs across a jet stream

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Dimensions of a jet stream

Jet Streaks (Wind Maxima): Along the axis of a jet stream there are centres of high speed winds called **Jet Streaks** — wind maxima. Weaker winds lie in between. Jet streaks retain their strength but move slower (by up to **25 kt**) than the jet stream itself. In a wavy jet they are located over or near the **ridge**.

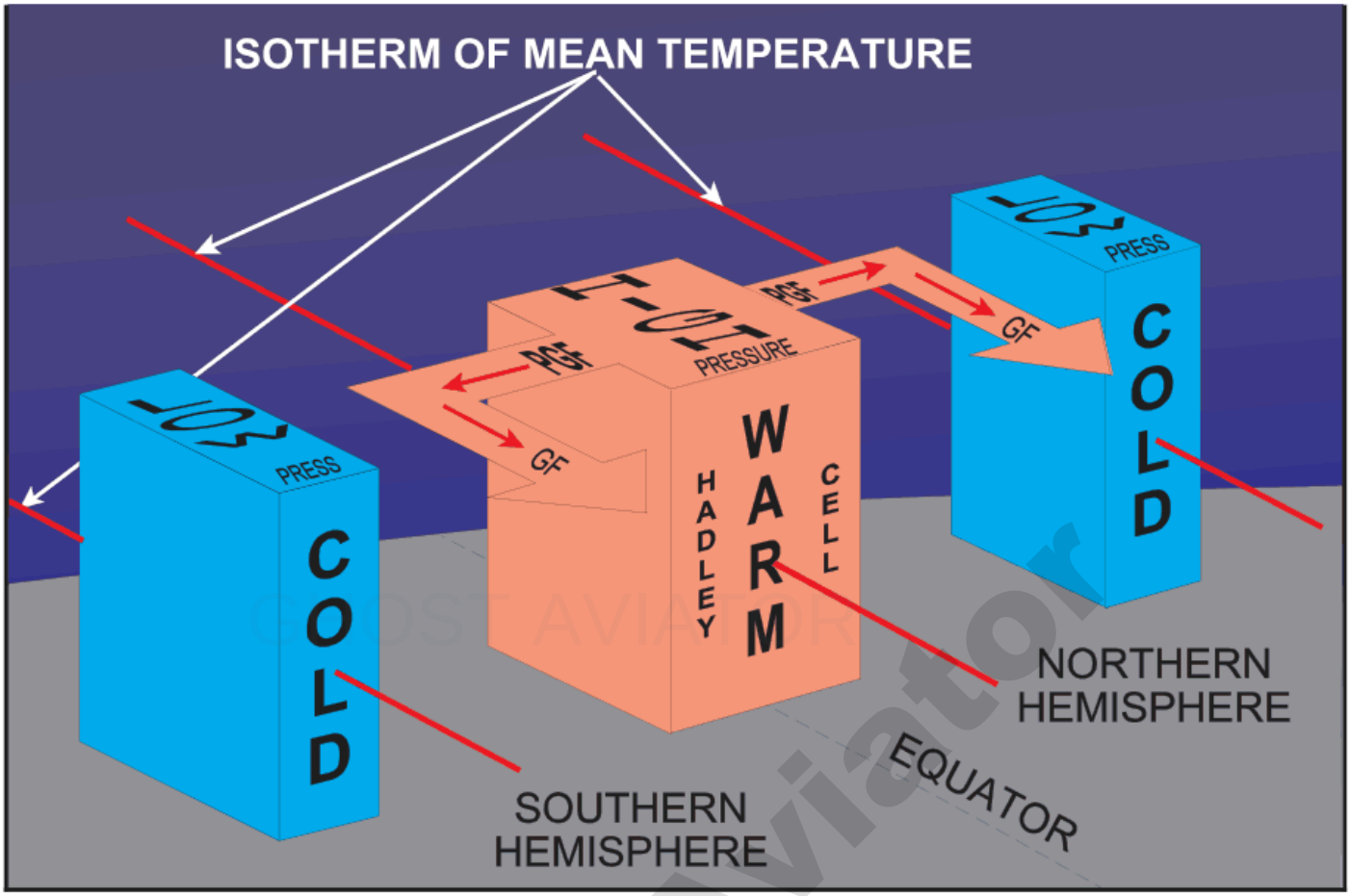
Wind speed along the axis is **NOT always uniform**. At certain places the wind speed reduces below 60 kt → constituting breaks in the jet stream. Sometimes jet streams split into two or more parts, combine, or flow in different directions.

🧠 **Path of maximum speed = the CORE (axis) of the jet stream.**
Jet Streaks = centres of high wind speed along the axis, located over/near ridges in wavy jets.

2. Types of Jet Streams

There are **four types** of jet streams in the troposphere:

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Sub-tropical jet streams

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Type	Symbol	Direction	Height / Pressure	Key Features
Arctic Jet Stream	AJ	Westerly	**7–8 km**, near polar tropopause	Over Arctic region, close to polar tropopause
Polar Front Jet	PFJ	Westerly (mainly); NWly or SWly on occasions	~ **250 hPa**, **9–12 km** (30,000–39,000 ft)	Also called mid-latitude jet. N America, Europe, Asia; SH = circles Antarctica. Migrates: winter → **30°N**; summer → **60°N**. Normal speed: **80–100 kt** in winters.
Sub-tropical Jet	STJ	**Westerly**	**10–16 km**, near mid-latitude tropopause break	Above subtropical high pressure belt. Winter position: S of **22°N**; Summer: N of **35°N** (weakens). Affects India: **Oct–May**. Speed: **100–215 kt** (up to 398 km/h).
Tropical Jet Stream	TJ	**Easterly**	~ **15–16 km**, ~ **150 hPa**	Also called Easterly Jet Stream. Prominent during monsoons, weak in winters. Speed: **60–80 kt** avg; max **150 kt**. Axis at **13°N** (Chennai lat.) over India in Jul–Aug. **Only over Asia and Africa.** NOT over Atlantic and Pacific.
Stratospheric Jet	SJ	Westerly (winter) / Easterly (summer)	Above **20 km**	In Arctic and Antarctic regions. Westerlies in winter; easterlies in summer.

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```
flowchart TB
  JS[JET STREAMS\nUpper Troposphere]
  JS --> AJ[Arctic Jet AJ\n7-8 km\nWesterly\nPolar regions]
  JS --> PFJ[Polar Front Jet PFJ\n9-12 km / 250 hPa\nWesterly\n30N winter → 60N summer\n80-100 kt]
  JS --> STJ[Sub-tropical Jet STJ\n10-16 km\nWESTERLY\n22N winter → 35N summer\n100-215 kt\nAffects India Oct-May]
  JS --> TJ[Tropical Jet TJ\n15-16 km / 150 hPa\nEASTERLY\nAxis 13N over India\nJun-Aug\nOnly Asia+Africa]
  JS --> SJ[Stratospheric Jet SJ\nAbove 20 km\nArctic/Antarctic regions]
```

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3. Sub-tropical Jet Stream (STJ) over India

STJ seasonal movement over India:

- Appears over N India in **October** after withdrawal of monsoon
- Progressively shifts southwards as winter progresses
- Mean position: roughly at **27°N** at height of **12 km**
- Reaches southernmost position at **22°N** in **February**
- Prevails over Indian subcontinent for about **8 months** (Oct to May)
- After May: shifts northwards and moves out of Indian subcontinent

STJ wind speeds over India:

- Mean wind speed: **100 kt**
- Maximum: **200 kt**
- Lowest (Oct and May): **60–70 kt**
- Highest (January): **100–120 kt**; on occasions core reaches **200 kt**
- Higher speeds more common in January and February than December
- Level of maximum wind: a **few km below the tropopause**

STJ structure:

- STJ splits due to Himalayas → one branch south, one north of Himalayas → recombine over China
- STJ has a **layered structure**: often two layers of maximum wind **to the S of the jet core**
- Jet stream strengthens by **10–15 kt** downstream, along axis from **Jodhpur to Guwahati**
- STJ located **3–4 km** below the tropopause in region where there is a break in tropical and middle tropopause at **300–200 hPa**

🧠 **STJ positions at a glance:**
Oct (arrival, 27°N, 12 km) → Feb (southernmost, 22°N) → May (exit northwards) → ABSENT Jun–Sep (monsoon season, TJ takes over)

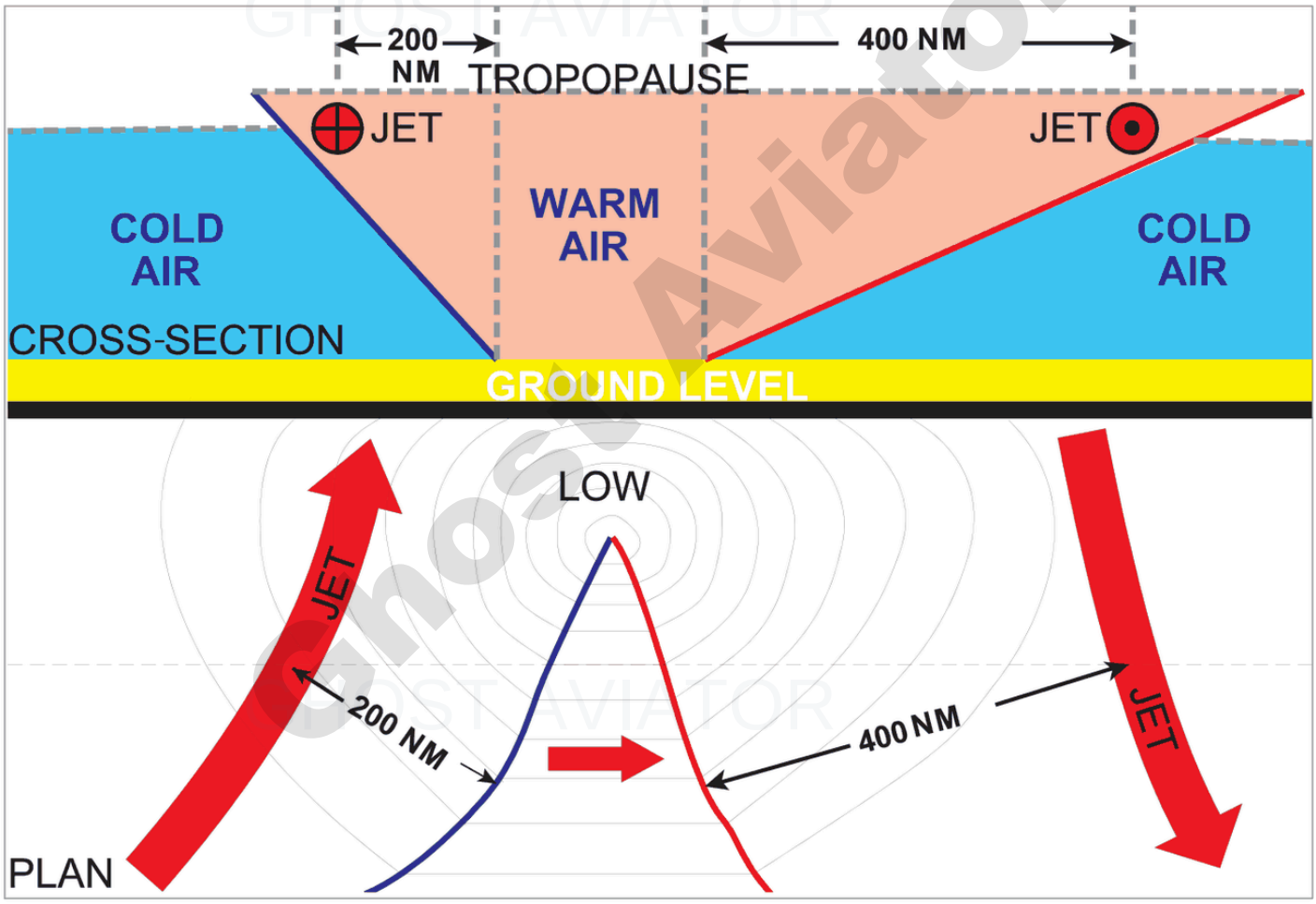
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4. Temperature Distribution & Wind Shear

STJ is caused by a concentration of horizontal temperature gradient below the jet and reversal above it.

- Surface to 300 hPa: horizontal temperature gradient from S to N (particularly at 500 and 300 hPa) → thermal wind builds up → jet maxima reached at **200 hPa**
- At 200 hPa: thermal gradient is slack and thereafter reverses
- **Lapse rates more to the S of the jetstream than to the N and above and below.**
- Near the STJ: temperature gradient is **very small or negligible**



Jet streams cross-section

Vertical shear: Wind speed changes rapidly above and below the jet core. For aviation hazards, distinguish this wind-speed profile from the location of strongest clear-air turbulence (CAT): a common CAT region is near or just below the core on the cold-air side, with a secondary region above.

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Horizontal Shear: In STJ, horizontal shear is more to the **north** than to the south of the jet core.

- At core level: winds weaken by about **35 kt/100 NM** to the south
- North side of front is steeper: about **100 kt/100 NM**
- Shear is more below the core and to the north (left looking downstream)

5. STJ and Weather

With the passage of a WD, STJ strengthens (**130–150 kt**) and moves south. Divergence in the left entrance and right exit of the jet core in combination with south → a low level system (low pressure) enhances its activity → may cause **severe TS**.

```
flowchart LR
  A[WD passes] --> B[STJ strengthens\n130-150 kt]
  B --> C[STJ moves south]
  C --> D[Divergence in left entrance\n+ right exit of jet core]
  D --> E[Low level system\nlow pressure enhanced]
  E --> F[SEVERE TS possible]
```

6. Tropical Jet Stream (TJ)

Also called: Easterly Jet Stream.

- Strong Easterlies blow to the S of **25°N** at **100 hPa** during monsoon
- **Found only over Asia and Africa; NOT over Atlantic and Pacific**
- During monsoon: **10–20 kt** Ely winds at 300 hPa, S of 27°N and Wly to N
- Prominent during monsoons; becomes very weak in winters
- Average core speed: **60–80 kt** ; maximum: **150 kt**
- TJ seen over India: **June to August**; disappears by **September**
- Axis: **13°N** (near Chennai/Madras latitude), at **150 hPa** , **15–16 km** height
- Level of maximum wind: **just below the tropopause**
- Wind shears in TJ much **less than STJ**

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Cause of TJ: Due to sun being N of 20°N in summer → southward decrease of temperature throughout the troposphere → affects the 20°N Tibetan Plateau. The Tibetan Plateau (being south in summers) causes lower level westerlies to change to easterlies above **500 hPa**. Easterlies strengthen with height up to **16 km**.

🧠 **TJ vs STJ:**
STJ = Westerly = Oct–May = 22–27°N = 12 km = 100–200 kt (strong)
TJ = Easterly = Jun–Aug = 13°N = 15–16 km = 60–150 kt (monsoon jet)
TJ = only Asia + Africa. STJ = global.

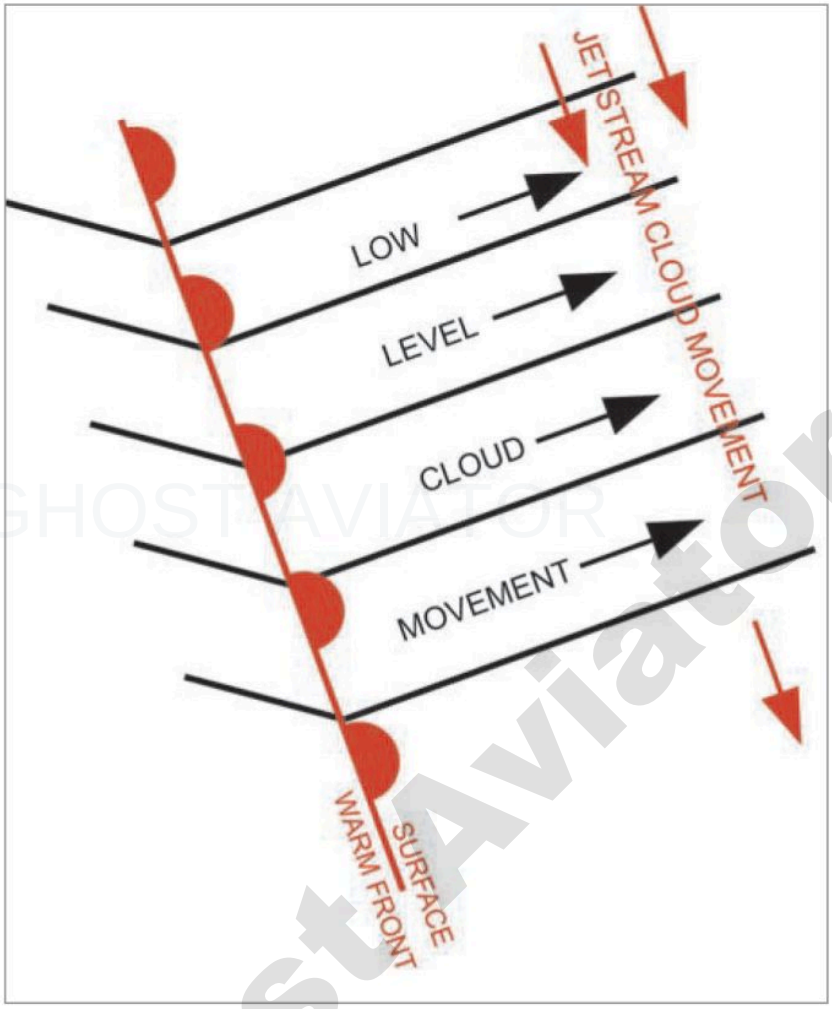
7. Low Level Jets (LLJ)

Very strong winds appear in the lower levels, between **1 km and 3 km** at many places in the sub-tropics. These are called **Low Level Jets (LLJ)**.

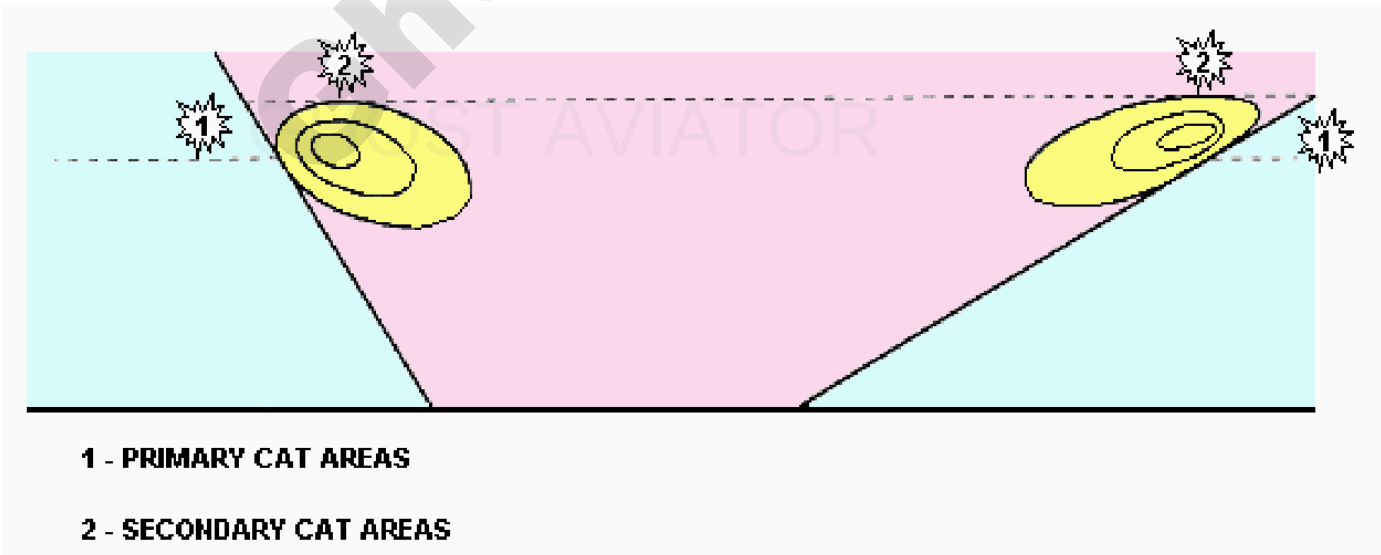
A prominent LLJ is noticed over the Indian Peninsula during SW monsoon when low level winds become very strong. This LLJ is probably a continuation of the jet stream off the **Somalia Coast in Africa**. The LLJ **enhances the monsoon activity** over the Indian Peninsula.

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8. Cause, Clouds & CAT associated with Jetstream

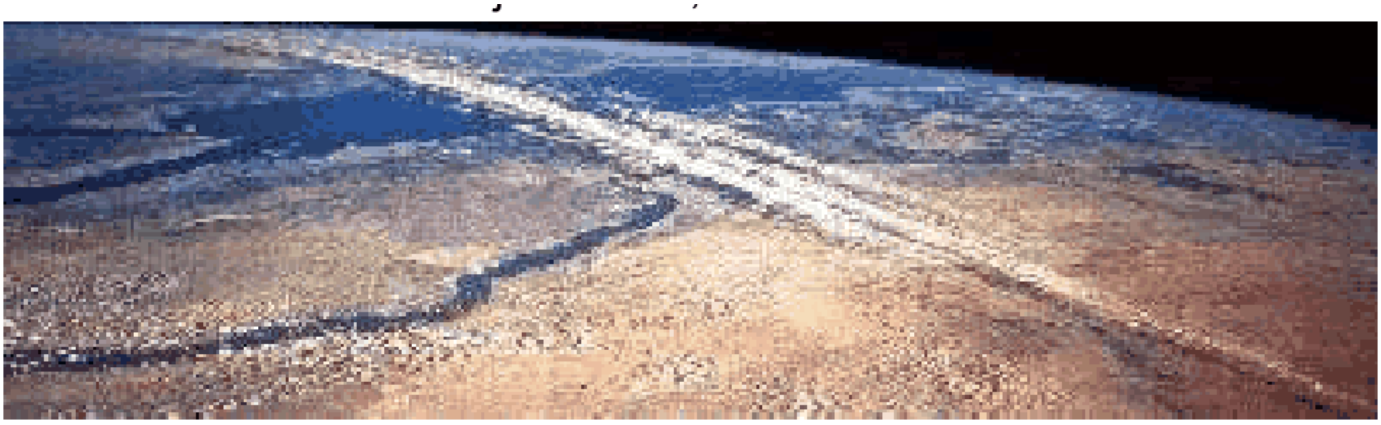


Cloud pattern at a jet stream



Clear air turbulence around the jet

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A jet stream seen from space

Cause

Jetstream is caused by the **large horizontal temperature gradient**. The larger the temperature difference between two air masses, the stronger is the jet stream.

Clouds Associated with Jetstream

- Extensive sheets of high clouds on the **warm side**, with sharply defined edges parallel to the jet axis
- **Ci clouds** more common on the **cold side**
- CC clouds on the cold side — tops invariably just below the tropopause
- Warm side cloud tops: a few thousand feet below the tropopause
- Roll type of Ci or CS may be seen along the length
- Where moisture is less: thin layer of haze on the warm side; clear on the cold side

Clear Air Turbulence (CAT)

CAT occurs around boundaries of Jetstream due to large vertical and horizontal wind shear. Also more in a developing and rapidly moving jet.

⚠ **Strongest CAT occurs near or just below the axis on the COLDER (low pressure) side, with a secondary maxima above the Jetstream.**

Looking downstream: CAT is to the **LEFT in the Northern Hemisphere**; to the right in Southern Hemisphere.

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CAT in a jetstream is most severe in:

1. Strong Winds
2. Curved Jets — the sharper the curve, the stronger the turbulence
3. On the leeward side and above mountain ranges

🧠 **CAT location mnemonic:** "Left of jet = Worst CAT" (Northern Hemisphere, looking downstream = cold/low pressure side). Pilot tip: if encountering CAT near jet, move to the RIGHT (warm side) when looking downstream.

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9. Quick Revision Summary

⚡ AMBER Quick Revision — Jet Streams

- **Jet stream lower limit:** **30 m/sec = 60 kt** (WMO arbitrary lower limit)
- **Vertical WS:** **5 m/s per km** | Depth often <5 km
- **Jet Streaks:** centres of high wind along axis; located over/near RIDGE in wavy jet; move slower than jet by up to 25 kt
- **Path of maximum speed = CORE (Axis)**
- **5 types:** Arctic (AJ), Polar Front (PFJ), Sub-tropical (STJ), Tropical (TJ), Stratospheric (SJ)
- **STJ = Westerly; TJ = Easterly** (key distinction)
- **STJ over India:** Oct (27°N, 12 km) → Feb (22°N, southernmost) → May exit | Mean 100 kt, Max 200 kt | Splits at Himalayas
- **STJ structure:** Two layers of max wind to the S of jet core | Strengthens 10–15 kt downstream (Jodhpur → Guwahati)
- **STJ + WD:** STJ strengthens to 130–150 kt + moves south → severe TS possible
- **TJ:** Easterly; Jun–Aug; 13°N (Chennai lat.); 15–16 km; 150 hPa; 60–80 kt (max 150 kt); only Asia + Africa; disappears by September
- **LLJ:** 1–3 km; SW monsoon; Somalia Coast origin; enhances monsoon over India
- **CAT strongest:** below axis on COLD side (left in NH); in strong winds, curved jets, leeward of mountains
- **Vertical shear:** Strong changes occur above and below the core. CAT is commonly strongest near or just below the cold-air side of the axis.
- **Horizontal shear:** More to the NORTH (left) of the jet
- **Clouds:** High clouds on WARM side; Ci more on COLD side

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10. Practice Q&A

Q1. The arbitrary lower limit of jet core velocity has been assigned by WMO as:
(a) 60 kt (b) 60 m/s (c) 70 m/s

✅ Answer: (a) 60 kt — The WMO lower limit of jet core speed is 60 kt, which is 30 m/s. Anything slower is not classed as a jet stream.

❌ (b) 60 m/s would be about 117 kt, far above the minimum. (c) 70 m/s is higher still.

💡 Jet stream: at least 60 kt (30 m/s).

Q2. Jet stream has:
(a) one maxima (b) one or more maxima (c) only two maxima

✅ Answer: (b) one or more maxima — The wind speed along a jet axis is not uniform: there are one or more maxima, called jet streaks.

❌ (a) and (c) set a fixed number of maxima, which the jet does not have.

💡 Jet streaks = the fast patches along the axis.

Q3. The vertical wind shear in a jet stream is:
(a) 5 m/sec/km (b) 6 m/sec/km (c) 4 m/sec/km

✅ **Answer: (a) 5 m/sec/km** — The text states: "The vertical WS is about 5 m/s per km." This is the key defining vertical wind shear value for jet streams.

❌ (b) 6 m/sec/km — not the stated value. ❌ (c) 4 m/sec/km — not the stated value.

💡 Jet stream vertical WS = **5 m/s per km**. Memorise this for DGCA.

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Q5. In a jet stream, the path of maximum speed is known as: (a) Core (b) Axis (c) Jet streak

✅ Answer: (b) Axis — The line along which the speed is greatest is the jet axis. The tube of fast air surrounding it is the core.

❌ (a) The core is the tubular volume around the axis. (c) A jet streak is a local maximum of speed along the axis.

💡 Axis = the path. Core = the tube. Streak = a fast patch.

Q6. Wind speed along the axis of a jet stream is always:
(a) same (b) uniform (c) not uniform

✅ **Answer: (c) not uniform** — The text explicitly states: "The wind speed along the axis is not always uniform. There are distinct maxima along the axis."

❌ (a)(b) Uniform/same — incorrect; jet streaks create distinct speed maxima along the axis.

💡 Jet speed = NOT uniform along the axis. Jet streaks create local maxima. Breaks occur where speed falls below 60 kt.

Q7. Along the axis of a jet stream there are centres of high speed winds, these are called:
(a) Ridge (b) Jet streaks (c) Between Trough and Ridge

✅ **Answer: (b) Jet streaks** — Centres of high speed wind along the jet axis = Jet Streaks (wind maxima). In a wavy jet, these are located over or near the ridge.

❌ (a) Ridge: the location of jet streaks in a wavy jet, not the name of the centres. ❌ (c) Not the correct terminology.

💡 Jet Streaks = wind maxima along jet axis. Located over/near ridges in wavy jets.

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Q8. Sub-tropical Jet Stream (STJ) is:
(a) Westerly (b) Easterly (c) Southerly

✅ Answer: (a) Westerly — The sub-tropical jet stream is a westerly jet. The tropical jet stream is the easterly one.

❌ (b) Easterly describes the Tropical Jet Stream (TJ), not the STJ. (c) Southerly is not a jet stream direction.

💡 STJ = westerly. TJ = easterly. Do not mix them.

Q11. The southern most position of STJ is in February:
(a) 22°N (b) 13°N (c) 17°N

✅ **Answer: (a) 22°N** — The text states: "It reaches its southern most position of 22°N in February."

❌ (b) 13°N: this is the latitude of the Tropical Jet Stream axis, not the STJ. ❌ (c) 17°N: not the stated value.

💡 STJ southernmost = **22°N in February**. TJ axis = **13°N (Chennai)**. Don't mix these up!

Q12. The STJ affects India from:
(a) Oct to May (b) Sep to Oct (c) Jun to Sep

✅ **Answer: (a) Oct to May** — STJ prevails over the Indian subcontinent for about 8 months from October to May. After May it shifts northwards and exits the Indian subcontinent.

❌ (b) Sep–Oct: too short; STJ arrives in October and stays till May. ❌ (c) Jun–Sep: this is the monsoon period when TJ (Easterly jet) prevails — STJ is absent then.

💡 STJ = Oct to May (8 months, non-monsoon). TJ = Jun to Aug (monsoon). When one is present, the other is absent over India.

Q13. STJ has a layered structure. There are often two layers of max wind to the ... of jet core:

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(a) S (b) N (c) SW

✅ **Answer: (a) S** — The text states: "There are often two layers of maximum wind, to the S of the jet core." This is the layered structure of the STJ.

❌ (b) N: horizontal shear is more to the N, but the two layers of max wind are to the S. ❌ (c) SW: not the stated direction.

💡 Two layers of max wind in STJ = **to the SOUTH of jet core**. Horizontal shear strongest = to the NORTH of jet core.

Q14. The STJ strengthens:
(a) Northwards (b) Upstream (c) Downstream

✅ **Answer: (c) Downstream** — "The jet stream strengthens by 10–15 kt downstream, along the axis from Jodhpur to Guwahati."

❌ (a) Northwards: lateral direction, not along jet axis. ❌ (b) Upstream: opposite of the text — weakens upstream, strengthens downstream.

💡 STJ strengthens DOWNSTREAM (W to E): Jodhpur → Guwahati, by **10–15 kt**. This helps forecasters — stronger STJ = more active weather over E India.

Q15. At and near the STJ the temperature gradient is very:
(a) Small (b) Large (c) Moderate

✅ **Answer: (a) Small** — "Near the STJ the temperature gradient is very small or negligible." The strong gradient exists BELOW the jet (up to 300 hPa); it weakens at the jet level and reverses above.

❌ (b) Large: the large gradient is BELOW the jet (causes the jet), not AT the jet level. ❌ (c) Moderate: the text says "very small or negligible."

💡 Temperature gradient: large BELOW jet (causes it) → very small/negligible AT jet level → reverses ABOVE jet. This is key to understanding jet stream physics.

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Q16. Vertical wind shear in STJ is greater:
(a) below (b) above (c) same above and below

✅ Answer: (a) below — The greatest vertical wind shear lies just below the jet core, on the cold-air side, with a secondary area of shear above the core.

❌ (b) Above: the secondary area above the core is weaker. (c) The shear is not the same above and below.

💡 CAT is worst just below the core on the cold side, with a secondary zone above it.

Q17. TJ prevails over Indian Peninsula from:
(a) May to Jun (b) Jun to Aug (c) Sep to Oct

✅ **Answer: (b) Jun to Aug** — The TJ (Easterly Jet Stream) is seen over India from June to August and disappears by September.

❌ (a) May–Jun: too early; TJ develops with monsoon onset in June. ❌ (c) Sep–Oct: TJ disappears by September; STJ returns in October.

💡 TJ = Jun → Aug → disappears Sep. STJ returns Oct. Perfect seasonal exchange over India.

Q18. TJ is located over India approximately at a height of:
(a) 11–12 km (b) 15–16 km (c) 8–10 km

✅ **Answer: (b) 15–16 km** — The text states: "It is at a mean height of about 15 to 16 km... The axis in July-August is near latitude 13°N (Chennai Lat.), at 150 hPa, over India."

❌ (a) 11–12 km: this is the height of the STJ (Sub-tropical Jet, at ~12 km). ❌ (c) 8–10 km: too low for a jet stream; PFJ is at 9–12 km.

💡 TJ height = **15–16 km at 150 hPa**. STJ height = **~12 km**. Don't mix heights!

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Q19. TJ is located over India approximately at a height of 15–16 km. It is in:
(a) July–Aug (b) Sep–Oct (c) Jun

✅ **Answer: (a) July–August** — "The axis in July-August is near latitude 13°N (Chennai Lat.), at 150 hPa." The TJ is active June–August but at maximum strength in July–August.

❌ (b) Sep–Oct: TJ disappears by Sep; STJ returns Oct. ❌ (c) Jun only: TJ starts in June but the axis specification is for Jul–Aug.

💡 TJ axis position (13°N, 15–16 km) is specified for July–August — peak monsoon period.

Q20. In the TJ the wind shears are much less than the STJ:
(a) True (b) False

✅ **Answer: (a) True** — The text states: "In the TJ the wind shears are much less than the STJ." The STJ has much stronger associated wind shears (and therefore stronger CAT potential) than the TJ.

❌ False: the text explicitly confirms TJ wind shears are less than STJ.

💡 TJ wind shear < STJ wind shear → Less CAT in TJ than STJ. The STJ is the more hazardous jet for aviation turbulence.

Q21. A jet stream can be recognised by: (a) High level dart (b) High Pressure (c) Streaks of Ci Lenticular clouds

✅ Answer: (c) Streaks of Ci Lenticular clouds — A jet stream shows up as long streaks and sheets of cirrus lined up along the axis.

❌ (a) and (b) do not identify a jet stream.

💡 Jet: streaky Ci along the axis.

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Q22. Flying at and near the STJ the aircraft will experience: (a) Wind from right (b) Increasing tail wind

✅ Answer: (b) Increasing tail wind — The sub-tropical jet is a westerly jet, so an aircraft flying along it with the flow (eastbound) gets an increasing tail wind.

❌ (a) Wind from the right applies to a track across the jet, not along it.

💡 Along the STJ eastbound: tail wind. Westbound: head wind.

Q23. Strongest CAT occurs near the jetstream on the:
(a) Warm side (b) Cold side (low pressure side) (c) On the jet axis itself

✅ **Answer: (b) Cold side (low pressure side)** — "Strongest CAT occurs near or just below the axis on colder (low pressure) side with a secondary maxima above the Jetstream." In the NH, this is to the LEFT looking downstream.

❌ (a) Warm side: some CAT but less severe. ❌ (c) On axis: maximum wind but not necessarily maximum CAT; shear zones are above/below and to the cold side.

💡 Strongest CAT = just below axis on COLD/LEFT side (NH). Secondary maximum = above the jet. Safest approach = from the warm/right side.

Q25. When and where tropical jet stream occurs: (a) All year along India (b) High level dart (c) In summers over SE Asia and Central Africa

✅ Answer: (c) In summers over SE Asia and Central Africa — The tropical (easterly) jet stream occurs in the summer monsoon months over Asia and Africa only, not over the Atlantic or the Pacific.

❌ (a) It is seasonal, not all year. (b) is not a place or season.

💡 TJ: June to August, Asia and Africa, easterly.

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11. Master Reference Tables

All Numerical Values — Chapter 15

Parameter	Value
WMO lower limit of jet stream velocity	**30 m/sec = 60 kt**
Jet stream vertical WS	**5 m/s per km**
Jet stream depth	Often less than **5 km**
Jet streak moves slower than jet by	Up to **25 kt**
PFJ height/pressure	**~250 hPa, 9–12 km (30,000–39,000 ft)**
PFJ normal speeds in winter	**80–100 kt**
PFJ winter migration	**30°N**
PFJ summer migration	**60°N**
STJ height	**10–16 km**
STJ mean position over India	**27°N, 12 km**
STJ southernmost position	**22°N in February**
STJ northward exit (summer)	North of **35°N**
STJ affects India (duration)	**8 months: Oct to May**
STJ mean wind speed over India	**100 kt**
STJ maximum wind speed	**200 kt**
STJ lowest speeds (Oct and May)	**60–70 kt**
STJ highest speeds (January)	**100–120 kt**
STJ speed range overall	**100 to over 215 kt (398 km/h)**
STJ level in tropopause region	**3–4 km** below tropopause; **300–200 hPa**

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STJ strengthens downstream by	**10–15 kt** (Jodhpur to Guwahati)
STJ strengthens with WD to	**130–150 kt**
STJ horizontal shear (north side)	About **100 kt/100 NM**
STJ horizontal shear (south side)	About **35 kt/100 NM**
TJ height over India	**15–16 km**
TJ pressure level	**150 hPa**
TJ axis latitude over India (Jul–Aug)	**13°N (Chennai latitude)**
TJ average core speed	**60–80 kt**
TJ maximum speed	**150 kt**
TJ active period over India	**June to August**
TJ disappears by	**September**
LLJ height	**1–3 km**
Breaks in jet stream (speed falls below)	**60 kt**
AJ height	**7–8 km**
SJ height	Above **20 km**

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Answer Key — Chapter 15

Q	A	Q	A	Q	A	Q	A
1	b*	2	a	3	b*	4	b
5	a	6	c	7	b	8	b*
9	b	10	c	11	a	12	b
13	a	14	c	15	a	16	c
17	b	18	a*	19	a	20	a
21	a	22	b	23	b	24	c
25	d						

* Items marked with * have apparent discrepancies between the answer key and the textbook content. Rely on the textbook explanation in the Q&A blocks above for conceptual understanding.

Mnemonics

- **Jet stream lower limit:** "Thirty metres per second = Sixty knots = JET"
- **STJ = W(esterly) / TJ = E(asterly)** — "Sub-tropical jets WEST, Tropical jets EAST"
- **STJ over India:** "OCT-27N → FEB-22N → MAY-exit" (shifts south in winter)
- **TJ = JAJA:** June–August, Asia–Africa only
- **STJ 8 months** (Oct–May) vs **TJ 3 months** (Jun–Aug)
- **Jet streaks:** Over RIDGES in wavy jet; move slower than jet by up to 25 kt
- **Strongest CAT = Cold side (LEFT in NH) just below jet axis**
- **Vertical wind shear:** Strong above and below the core; distinguish the wind profile from the common CAT region near or below the cold-air side.
- **STJ + WD = 130–150 kt + South shift → Severe TS**
- **LLJ = 1–3 km + Somalia origin + Enhances monsoon**
- **TJ axis over India = 13°N (Chennai) = 15–16 km = 150 hPa**
- **STJ axis over India = 27°N (mean) → 22°N (Feb southernmost) = 12 km**
- **CAT worst in: Strong winds + Curved jet + Leeward side of mountains**

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